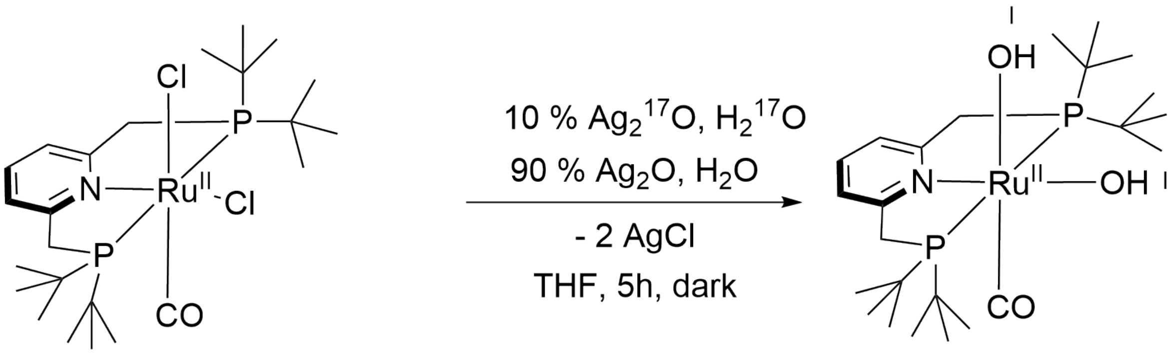
AE-460: Synthesis of 17O labled [Ru(CO)(OH)2(PNPtBu)]

Date: 2025-03-12
Tags: Dihydroxo Ag2O PNP P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)] NMR AE Future reference analytics 1H 31P 17O
Category: Two-photon
Status: Done
Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: $C_{24}H_{43}Cl_2NOP_2Ru$
Molecular Weight: 595,53152

Chemical Formula: $C_{24}H_{45}NO_3P_2Ru$
Molecular Weight: 558,64552

$[RuCl_2(CO)(PNN)]$

Silver Oxide

$[Ru(CO)(OH)_2(PNN)]$

Molecular Weight:
231.74 g/mol

Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Reagents

Name	CAS-Number/Expermient-Number	Amount [mmol]	Equivalents	$Mass_{theo}$ [mg]	$Mass_{exp}$ [mg]	Molar mass [g/mol]	$Volume_{theo}$ [ml]	$Volume_{exp}$ [ml]	Density [g/mL]
[RuCl2(CO)(PNP{tBu})]	Two-photon - KRA-DS-02: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)] I	0.118	1.00	70.47	70.76	595.53	/	/	/
$Ag_2{}^{17}O$ (10 % 17O)	Two-photon - AE-450: Synthesis of Ag2(17)O from AgNO3 and H2(17)O	0.727	6.14	168.38	167.60	231.74	/	/	/

$H_2^{17}O$ (1o % 17O)	Two-photon - AE-459: Degassing of H2(17)O	27.69	234	499	/	18.02	0.500	0.50	0.998
THF	Prep work - AE-396: Degassing of THF	/	/	/	/	/	3.5	3.6	/

Work-up Reagents

Name	CAS Number / Expermient-Number	Inventory number	$Mass_{exp}$ [g]	Volume [ml]	Concentration [M]
THF	Prep work - AE-396: Degassing of THF	/	/	7.5	/
n-heptane	AE-315: Degassing of n-Heptane	/	/	14 + 3	/
MeCN-d3	KRA-056: Degassing of MeCN-d3	/	/	1	/

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol Protocol - Schlenk Technique)

Due to the instability of Ag_2O and Dihydroxo to the light, the procedure was done mostly in the dark, by either wrapping the flasks containing either or both in Aluminium foil or by tuning off the light in the lab and fumehood

Date	Time	Step	Fotos	Observations
12.03		A 25 mL brown glas Schlenk flask was prepared, by heating under vacuum	/	/
	9:10	$[RuCl_2(CO)(PNP)]$ and Ag2(17)O was weighted in air into the reaction schlenk flask.	/	/
		The light in the fumehood was turned out	/	/
	9:25	The flask was set under argon	/	/
	9:42	THF (3.6 mL) was added to the reaction schlenk flask.	/	/
	9:45	H2(17)O (0.50 mL) was added to the reaction schlenk flask	/	/
	9:45 - 14:45	The reaction mixture was stirred for 5 h at 500 rpm. During this the flask was wrapped with aluminium foil	/	/
		In the meantime a 100 mL Schlenk flask was prepared	/	/
	14:45 - 14:55	The reaction mixture was allowed to settle	/	/
		Work-up:		
	14:55	The obtained suspension was filtered through a PTFE syringe filter (pore diameter: 0.2 µm) in the new 100 mL flask	after filt through PTFE.jpg	purplelish solution

	15:30 - 15:32	The solvents of the filtrate were removed under reduced pressure, for this no water bath was used	/	The mixture solidified
	15:33	A water bath (rt) was was put under the flask, while keeping the flask under an reduced pressure	/	/
	15:40 - 16:18	The obtained solid was dried under reduced pressure	after drying.jpg	**AE-460-0**, brown film
		The liquid in the cooling trapped was transferred into a 10 mL snap on cap vial. It was stored under air in the SSC at rt	liquid from drying.jpg (mislabeld as -2 in Photo)	**AE-460-1**
	16:20	**AE-460-0** was stored in the dark at -20 °C under argon	/	/
		From now on **AE-460-0**	next morning.jpg	no change
13.03	10:10	To the solid THF (7.5 mL) was added und the obtained mixture was stirred at 500 rpm. In between the flask was shaken for approx. 30 s.	in THF.jpg	light brwon, pinkish suspension
	10:17	n-heptane (14 mL) was added to the mixture under stirring	addition of heptane.jpg setteled.jpg	pinkish solution, yellowish solid
	10:27 - 10:54	The mixture was filtered through a pore 4 frit according to Protocol - Filtration with frit technique into a 100 mL-Schlenk	during filt.jpg	no change
	10:54 - 11:00	The solid was washed with n-heptane (3 * 1.0 mL)	product.jpg	brwonish solid
	11:02 - 13:05	The solid was dried under reduced pressure	/	**AE-460-2**
		2 10 mL Schlenk flask (1 with a stir bar, 1 with a 2 mL screw cap vial) were prepared	/	/
		From now on **AE-460-2:**		
	13:30 - 13:40	The solid was transfferred into a 2 mL vial in a 10 mL Schlenk flask in air, thereby it was weighed	/	/
		During this an aliqoute (8.97 mg) was transfferred into the other 10 mL Schlenk flask	/	/
		Both flask were set under argon	/	/
		AE-460-2 was stored in Equipment - Freezer Lab E004 CEEC II in the dark at -20 °C under Ar	/	/
	15:28	To the aliquote MeCN-d3 (1 mL was added)	solution in MeCN-d3.jpg	orange brown solution
	- 15:32	The obtained solution was stirred at 600	/	/
	15:33	From this an NMR sample was prepared according to Protocol - Preparation of NMR Sample during this the solution was filtered through an PTFE syringe filter (pore diameter: 0.2 µm)	/	/

	15:45	The NMR was subimtted for an 1H NMR	NMR sample.jpg	**AE-460-2-1** yellow orange solution
14.03	12:00	The same NMR tube was submitted for an 17O NMR	/	**AE-460-2-2**
19.03	11:10	The same NMR tube was submitted for several NMR experiments		**AE-460-2-3**
26.03	16:00	The same NMR tube was submitted for 13C NMR experiment on 500 MHz device	AE-460-2-4.jpg	**AE-460-2-4** very small amount of solid observed at the wall, otherwise no change, by eyesight should be still ok

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
13.03	16:01	AE-460-2-1	NMR 1H	ZAF 300 MHz OA	MeCN-d3	AE-460-2-1.zip	AE-460-2-1_10.nmrium	Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (**AE-382-2**)	As expected. Very few impurties observed.
14.03	12:46	AE-460-2-2	NMR 1H, 31P{1H}, 17O{1H}	IAAC 400 MHz	MeCN-d3	AE-460-2-2.zip	AE-460-2-2_10.nmrium	Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (**AE-382-2**)	In 1H and 31P as expected. Very few impurities observed. In 17O: 3 signals observed. As expected all broad singulets. Shifts (intensity): 19.8 (43 %), 94.7 (47 %) and 244.0 (9.5 %) ppm. Main signals would roughly fit to OH groups (expected shift range for alkohols: very approx. -50 to 100 ppm according to https://chem.ch.huji.ac.il/nmr/techniques/1d/row2/o.html) Signal at 244 ppm: no good idea, shift qould fit to peroxide, carbonate or carboxylic acid according to https://chem.ch.huji.ac.il/nmr/techniques/1d/row2/o.html
19.03	15:36	AE-460-2-3	NMR 1H, 13C{1H}, 31P{1H}, 13C{1H}-APT, 1H/1H-COSY, 1H/13C{1H}-HMBC, 1H/13C{1H}-HSQC, 1H/31P{1H}-HMBC	IAAC 400 MHz I	MeCN-d3	AE-460-2-3-all-spectra.zip AE-460-2-3-1H.zip AE-460-2-3-31P.zip AE-460-2-3-1H/31P-HMBC.zip AE-460-2-3-13C.zip AE-460-2-3-1H/13C-HSQC.zip AE-460-2-3-1H/13C-HMBC.zip AE-460-2-3-1H/1H-COSY.zip AE-460-2-3-13C-APT.zip	AE-460-2-3-1H-13C-31P.nmrium AE-460-2-3-1H-13C-31P-comparison.nmrium AE-460-2-3-31P-HMBC.nmrium AE-460-2-3-13C-HSQC.nmrium AE-460-2-3-13C-HMBC.nmrium AE-460-2-3-1H-COSY.nmrium	**AE-460-2-2**	1H, 31P as expected, all signals identified 13C: Evaluation pending, some tripletts (?) observed around 38 ppm 31P-HMBC: all cross signals observed 13C-HSQC: all cross singals observed 13C-HMBC: evaluaion pending, broad couplings 1H-COSY: Cross signals observed for aromatic signals and for methylene signals
28.03	12:56	AE-460-2-4	NMR 1H, 13C	IOMC 500 MHz	MeCN-d3	AE-460-2-4.zip	AE-460-2-4_101.nmrium	**AE-460-2-2** **AE-460-2-3**	13C: all signals observed, signals of CH2 and Cq of tBu no triplett observed 1H: mainly as expected, but for tBu groups more complex coupling patterns observed. Looks like weak douplet of tripletts

Product characterization and results

Sample	Mass [mg]	Purity	$Mass_{pure}$ [mg]	Amount [µmol]	Yield [%]	Description	Storage location
AE-460-2	26.51	>98 %	/	47.5	40	dried product.jpg light greenish gray solid	Equipment - Freezer Lab E004 CEEC II

Results	High yield obtained. Singlas observed in 17O, two of the 3 would fit to dihydroxo groups

Future recommendations

Old procedure	Problem	Suggested new procedure

/	/	always use self made silver(I) oxide

Linked experiments

- AEI-010: Synthesis of [Ru(CO)(OH)2(PNN)] using Ag2O
- JSC-KS-04: Degassing of n-Heptane
- AEI-024: Synthesis of [Ru(CO)(OH)2(NNN)] with 2,6-bis(N,N-diethylaminomethyl) using Ag2O
- AEI-026: Capillary production
- AEI-048: Synthesis of [Ru(CO)(OH)2(NNN)] with 2,6-bis(N,N-diisopropylaminomethyl) using Ag2O
- AEI-064: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O
- AEI-073: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O
- AE-101: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]
- AE-102: Degassing of THF
- AE-105: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O
- AE-108: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]
- AE-114: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O
- AE-121: Degassing of n-Heptane
- AE-145: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O
- AE-152: Degassing of MeCN
- AE-157: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]
- AE-167: Irradiation of PhPDA (365 nm + white LED, gas phase measurement), AE-151, 1.5 mg/mg SDS, 2 mg/mL PhPDA
- KRA-056: Degassing of MeCN-d3
- AE-198: Synthesis of [Ru(CO)(OH)2(PNP{tBu})] with Ag2O with different reaction times (1, 2, 5, 24 h)

- AE-KRA-204: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

- AE-208: Degassing of Milli-Q water

- KRA-080: Degassing of THF

- KRA-104: Degassing of Milli-Q water

- AE-230: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-232: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-238: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

- AE-246: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-238, 97 % pure) 365 nm and white

- AE-252: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

- AE-261: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)]

- AE-281: Stability test of [Ru(CO)(OH)2(PNP{tBu})] (AE-252) under air and afterwards irradiation with 365 nm and white

- AE-310: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

- AE-315: Degassing of n-Heptane

Prep work - AE-396: Degassing of THF

Prep work - AE-442: Degassing of Milli-Q water

Two-photon - AE-382: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Two-photon - AE-450: Synthesis of Ag2(17)O from AgNO3 and H2(17)O

Two-photon - KRA-DS-02: Synthesis of [RuCl2(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl2(CO)(p-cymene)] I

Two-photon - AE-456: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Two-photon - AE-459: Degassing of H2(17)O

Linked resources

Equipment - Freezer Lab E004 CEEC II

Protocol - Preparation of NMR Sample

Protocol - Filtration with frit technique

Protocol - Schlenk Technique

Protocol - Preparation of capillary for H2O-NMR

Attached files

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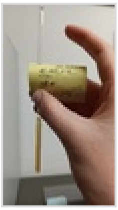

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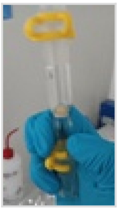

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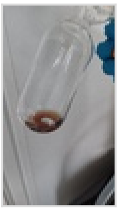

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Comment

On 2025-03-18 11:42:16 Jacob Schneidewind wrote:
Notes:
* Fixing link to H2[17O] water preparation, specifying that water and Ag2O are 10% 17O labelled
* 9:25 deleting duplication
* Putting dashes into empty fields (photos or observations etc.)
* Including information on appearance of reactants (Ru(PNP) and Ag2O etc.)
* Making "light turned off" in grey
* Specifying that stirring was turned off at 14:45
* 14:55: fixing uM to um
* 15:33: specifying room temperature water bath
* at 14:55: writing work-up procedure
* 13.03, 13:30: specifying that solid was transferred in air, and specifying that at this point the sample was weighed, and referring to AE-460-2 instead of "the solid"
* Adding MeCN-d3 to work-up reagents
* Having heading for results
* Adding future recommendation: using freshly prepared Ag2O

Unique eLabID: 20250310-9e0bbe50a767eac9cc87147abd8cc955b206a975
Link: https://elab.water-splitting.org/experiments.php?mode=view&id=1857